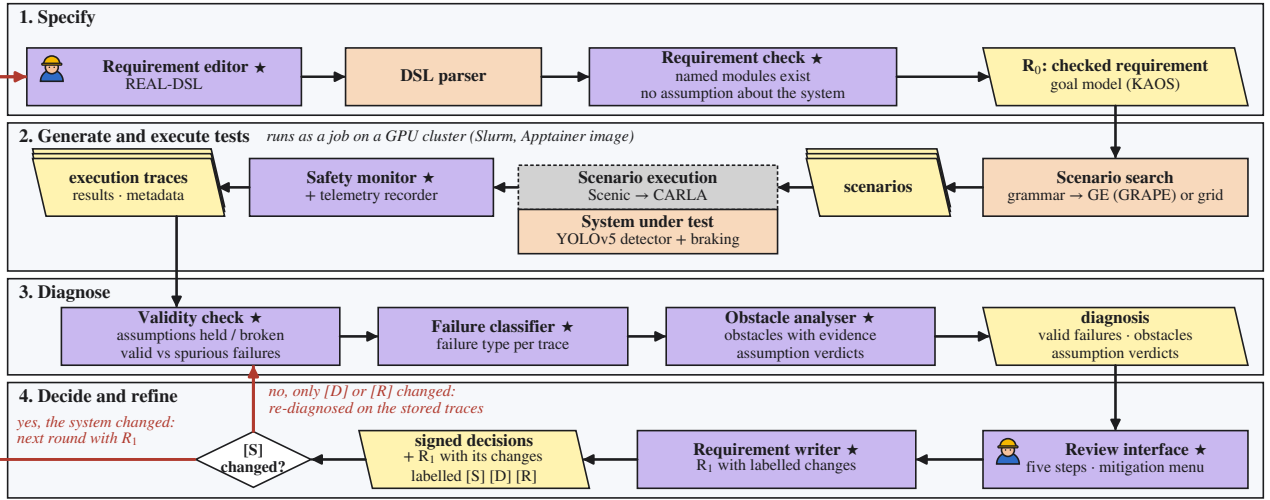
1. Specify
Requirement editor ★
REAL-DSL
DSL parser
Requirement check ★
named modules exist
no assumption about the system
R0: checked requirement
goal model (KAOS)
2. Generate and execute tests
runs as a job on a GPU cluster (Slurm, Apptainer image)
execution traces
results · metadata
Safety monitor ★
+ telemetry recorder
Scenario execution
Scenic → CARLA
System under test
YOLOv5 detector + braking
scenarios
Scenario search
grammar → GE (GRAPE) or grid
3. Diagnose
Validity check ★
assumptions held / broken
valid vs spurious failures
Failure classifier ★
failure type per trace
Obstacle analyser ★
obstacles with evidence
assumption verdicts
diagnosis
valid failures · obstacles
assumption verdicts
4. Decide and refine
yes, the system changed:
next round with R1
no, only [D] or [R] changed:
re-diagnosed on the stored traces
[S]
changed?
signed decisions
+ R1 with its changes
labelled [S] [D] [R]
Requirement writer ★
R1 with labelled changes
Review interface ★
five steps · mitigation menu
component (process)
artefact (input / output)
a set of many
decision
stage
operated by the engineer; the rest is automatic
new in this paper
reused from the REAL framework package
existing tool
data flow
loop back